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1 C:\Users\sumer\Desktop\Projects\Python\test\venv\Scripts\python
  .exe C:\Users\sumer\Desktop\Projects\Python\test\svm.py
2 Classification report for all hyperparameter combinations:
3 Hyperparameters: {'C': 0.1, 'kernel': 'linear'}
4 Mean test score: 0.809 (std: 0.005)
5 Classification report:
6           precision    recall  f1-score   support
7
8      0           0.84      0.96      0.89       7040
9      1           0.67      0.33      0.44       1960
10
11     accuracy                0.82       9000
12    macro avg           0.75      0.64      0.67       9000
13 weighted avg           0.80      0.82      0.79       9000
14
15 =====
16 Hyperparameters: {'C': 0.1, 'kernel': 'rbf'}
17 Mean test score: 0.815 (std: 0.004)
18 Classification report:
19           precision    recall  f1-score   support
20
21      0           0.84      0.96      0.89       7040
22      1           0.67      0.33      0.44       1960
23
24     accuracy                0.82       9000
25    macro avg           0.75      0.64      0.67       9000
26 weighted avg           0.80      0.82      0.79       9000
27
28 =====
29 Hyperparameters: {'C': 0.1, 'kernel': 'poly'}
30 Mean test score: 0.794 (std: 0.002)
31 Classification report:
32           precision    recall  f1-score   support
33
34      0           0.84      0.96      0.89       7040
35      1           0.67      0.33      0.44       1960
36
37     accuracy                0.82       9000
38    macro avg           0.75      0.64      0.67       9000
39 weighted avg           0.80      0.82      0.79       9000
40
41 =====
42 Hyperparameters: {'C': 1, 'kernel': 'linear'}
43 Mean test score: 0.809 (std: 0.004)
44 Classification report:

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45           precision    recall  f1-score   support
46
47      0           0.84      0.96      0.89       7040
48      1           0.67      0.33      0.44       1960
49
50      accuracy                0.82       9000
51      macro avg           0.75      0.64      0.67       9000
52      weighted avg        0.80      0.82      0.79       9000
53
54 =====
55      =====
56 Hyperparameters: {'C': 1, 'kernel': 'rbf'}
57 Mean test score: 0.819 (std: 0.002)
58 Classification report:
59           precision    recall  f1-score   support
60
61      0           0.84      0.96      0.89       7040
62      1           0.67      0.33      0.44       1960
63
64      accuracy                0.82       9000
65      macro avg           0.75      0.64      0.67       9000
66      weighted avg        0.80      0.82      0.79       9000
67
68 =====
69      =====
70 Hyperparameters: {'C': 1, 'kernel': 'poly'}
71 Mean test score: 0.804 (std: 0.003)
72 Classification report:
73           precision    recall  f1-score   support
74
75      0           0.84      0.96      0.89       7040
76      1           0.67      0.33      0.44       1960
77
78      accuracy                0.82       9000
79      macro avg           0.75      0.64      0.67       9000
80      weighted avg        0.80      0.82      0.79       9000
81
82 =====
83      =====
84 Best hyperparameters found: {'C': 1, 'kernel': 'rbf'}
85 SVM Classification Accuracy: 0.8186666666666667
86
87 Process finished with exit code 0
88
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